PHYS 143 – Mechanics and Thermal Physics

Introduction: Physical Quantities and Measurement – Tutorial Sheet

Hope Donglo, Office 24, Department of Physics, University of Ghana

Use I SEE for every problem: identify, set up, execute, evaluate. Predict first, write each equation in symbols before you substitute, carry the units through every line, and keep one extra digit until the last line. Give final answers with the correct number of significant figures. Part A has the problems, Part B has the worked solutions, and Part C has multiple-choice practice in the exam style. Try each problem before you read its solution.

A. Practice problems (easier to harder)

A1. Converting a speed

Convert $72\,\mathrm{km/h}$ to $\mathrm{m/s}$, and $20\,\mathrm{m/s}$ to $\mathrm{km/h}$. Predict first which number is larger in each case.

A2. Speed of sound

The speed of sound in air is about $3.4 \times 10^2\,\mathrm{m/s}$. Express it in $\mathrm{km/h}$.

A3. Converting a volume

A tank holds $2.5 \times 10^3\,\mathrm{cm}^3$. Express this in m^3 and in litres, using 1 litre $= 10^{-3}\,\mathrm{m}^3$.

A4. Cubing a prefix

Show that $1\,\mathrm{km}^3 = 10^9\,\mathrm{m}^3$ and that $1\,\mathrm{mm}^3 = 10^{-9}\,\mathrm{m}^3$.

A5. Unit consistency

Which of these equations is dimensionally consistent? (a) $v = v_0 + at$; (b) $x = v_0 t + at$; (c) $v^2 = v_0^2 + 2ax$. Here x is a length, v a speed, t a time and a an acceleration in $\mathrm{m/s}^2$.

A6. Significant figures

(a) State the number of significant figures in $0.0450\,\mathrm{m}$, $1.20 \times 10^3\,\mathrm{g}$, $2.003\,\mathrm{s}$ and $6.0 \times 10^{-4}\,\mathrm{m}^3$.
(b) Evaluate with the correct number of significant figures: 12.5×3.2; $18.52 + 4.6$; $7.25 \div 2.0$; $1.2 \times 10^3 + 45$.

A7. Uncertainty of a wire

A micrometer caliper gives the diameter of a wire as $(2.91 \pm 0.01)\,\mathrm{mm}$. Find the fractional and percent uncertainty, and the cross-sectional area $A = \pi d^2/4$ with the correct number of significant figures.

A8. Accurate or precise?

A standard mass is exactly $1.000\,\mathrm{kg}$. Student X reads 1.0213, 1.0215 and $1.0214\,\mathrm{kg}$. Student Y reads 0.9, 1.1 and $1.0\,\mathrm{kg}$. Decide who is more accurate and who is more precise.

A9. Order-of-magnitude estimate

Estimate the number of heartbeats in a lifetime of 70 years, taking 70 beats per minute. Give the answer to one significant figure and state its order of magnitude.

A10. Density of a block

A metal block measures $2.50\,\mathrm{cm} \times 4.0\,\mathrm{cm} \times 12.35\,\mathrm{cm}$ and has mass $1.20 \times 10^{3}\,\mathrm{g}$. Find its density in $\mathrm{g/cm^{3}}$ and in $\mathrm{kg/m^{3}}$ with the correct number of significant figures.

B. Worked solutions to A1 to A10

A1. Converting a speed

Predict: the number in m/s is smaller than the number in km/h, because the hour is much longer than the second.
IDENTIFY: convert km/h to m/s and back. Use $1\,\mathrm{km} = 1000\,\mathrm{m}$ and $1\,\mathrm{h} = 3600\,\mathrm{s}$.
SET UP: $v = v_{\text{given}} \times \dfrac{1000\,\mathrm{m}}{1\,\mathrm{km}} \times \dfrac{1\,\mathrm{h}}{3600\,\mathrm{s}}$, and the inverse factors for the reverse conversion.
EXECUTE:
$72\,\mathrm{km/h} = 72 \times \dfrac{1000\,\mathrm{m}}{3600\,\mathrm{s}} = \boxed{20\,\mathrm{m/s}}$.
$20\,\mathrm{m/s} = 20 \times \dfrac{1\,\mathrm{km}}{1000\,\mathrm{m}} \times \dfrac{3600\,\mathrm{s}}{1\,\mathrm{h}} = \boxed{72\,\mathrm{km/h}}$.
EVALUATE: the m/s number is smaller, as predicted. Both answers fit the rule: divide km/h by 3.6 to get m/s.

A2. Speed of sound

Predict: about 3.6 times the m/s value, so a little over $1000\,\mathrm{km/h}$.
IDENTIFY: convert $v = 3.4 \times 10^2\,\mathrm{m/s}$ (two significant figures) to km/h.
SET UP: $v = v_{\text{given}} \times \dfrac{1\,\mathrm{km}}{1000\,\mathrm{m}} \times \dfrac{3600\,\mathrm{s}}{1\,\mathrm{h}}$.
EXECUTE: $v = 3.4 \times 10^2 \times \dfrac{3600}{1000}\,\mathrm{km/h} = 1224\,\mathrm{km/h}$. Keep two significant figures: $\boxed{1.2 \times 10^3\,\mathrm{km/h}}$.
EVALUATE: consistent with the prediction. Writing $1224\,\mathrm{km/h}$ would claim four significant figures, which the data do not support.

A3. Converting a volume

Predict: a few thousand cm^3 is only a few litres, and the m^3 number is tiny.
IDENTIFY: convert cm^3 to m^3 and to litres.
SET UP: $1\,\mathrm{cm} = 10^{-2}\,\mathrm{m}$, so $1\,\mathrm{cm}^3 = (10^{-2}\,\mathrm{m})^3 = 10^{-6}\,\mathrm{m}^3$. Also 1 litre $= 10^{-3}\,\mathrm{m}^3$.
EXECUTE:
$V = 2.5 \times 10^3\,\mathrm{cm}^3 \times \dfrac{10^{-6}\,\mathrm{m}^3}{1\,\mathrm{cm}^3} = \boxed{2.5 \times 10^{-3}\,\mathrm{m}^3}$.
$V = 2.5 \times 10^{-3}\,\mathrm{m}^3 \times \dfrac{1\text{ litre}}{10^{-3}\,\mathrm{m}^3} = \boxed{2.5\text{ litres}}$.
EVALUATE: $10^3\,\mathrm{cm}^3$ is one litre, so $2.5 \times 10^3\,\mathrm{cm}^3$ is 2.5 litres. The two routes agree.

A4. Cubing a prefix

IDENTIFY: a volume unit is a length unit cubed, so cube the length factor.
SET UP: $1\,\mathrm{km} = 10^3\,\mathrm{m}$ and $1\,\mathrm{mm} = 10^{-3}\,\mathrm{m}$.
EXECUTE: $1\,\mathrm{km}^3 = (10^3\,\mathrm{m})^3 = 10^9\,\mathrm{m}^3$. $1\,\mathrm{mm}^3 = (10^{-3}\,\mathrm{m})^3 = 10^{-9}\,\mathrm{m}^3$.
EVALUATE: a cubic millimetre is a very small volume and a cubic kilometre a very large one, so the signs of the exponents are right. The exponent is multiplied by 3, not added.

A5. Unit consistency

IDENTIFY: two terms may be added or equated only if they have the same units. Check each term.
SET UP: treat units as algebraic symbols.

EXECUTE:
(a) $\mathrm{m/s} = \mathrm{m/s} + (\mathrm{m/s^2})(\mathrm{s}) = \mathrm{m/s} + \mathrm{m/s}$. Consistent.
(b) $\mathrm{m} = (\mathrm{m/s})(\mathrm{s}) + (\mathrm{m/s^2})(\mathrm{s}) = \mathrm{m} + \mathrm{m/s}$. Not consistent, because a length cannot be added to a speed.
(c) $\mathrm{m^2/s^2} = \mathrm{m^2/s^2} + (\mathrm{m/s^2})(\mathrm{m}) = \mathrm{m^2/s^2} + \mathrm{m^2/s^2}$. Consistent.
EVALUATE: answer: (a) and (c). Consistency is necessary but not sufficient: a consistent equation can still be wrong by a pure number.

A6. Significant figures

IDENTIFY: counting rules for part (a); the multiplication and addition rules for part (b).
SET UP: leading zeros do not count; zeros after the decimal point, or between non-zero digits, count; in addition and subtraction the decimal places decide, in multiplication and division the number of significant figures.
EXECUTE: (a) $0.0450\,\mathrm{m}$: $\boxed{3}$. $1.20 \times 10^3\,\mathrm{g}$: $\boxed{3}$. $2.003\,\mathrm{s}$: $\boxed{4}$. $6.0 \times 10^{-4}\,\mathrm{m^3}$: $\boxed{2}$.
(b) $12.5 \times 3.2 = 40.0$; the factor 3.2 has two figures: $\boxed{4.0 \times 10^1}$.
$18.52 + 4.6 = 23.12$; 4.6 has one decimal place: $\boxed{23.1}$.
$7.25 \div 2.0 = 3.625$; 2.0 has two figures: $\boxed{3.6}$.
$1.2 \times 10^3 + 45 = 1245$; 1.2×10^3 is uncertain to about 100: $\boxed{1.2 \times 10^3}$.
EVALUATE: the last two results show that sums and products follow different rules: for sums the position of the last reliable digit decides, not the count of figures.

A7. Uncertainty of a wire

Predict: the percent uncertainty of a $2.91\,\mathrm{mm}$ measurement with a $0.01\,\mathrm{mm}$ uncertainty is a few tenths of a percent.
IDENTIFY: fractional uncertainty, percent uncertainty and $A = \pi d^2/4$.
SET UP: fractional uncertainty $= \dfrac{\text{uncertainty}}{\text{measured value}}$; percent uncertainty is 100 times that.
EXECUTE:
$\dfrac{0.01\,\mathrm{mm}}{2.91\,\mathrm{mm}} = 0.00344 \approx \boxed{0.003}$; percent uncertainty $\approx \boxed{0.3\%}$.
$A = \dfrac{\pi(2.91\,\mathrm{mm})^2}{4} = \dfrac{\pi\,(8.4681\,\mathrm{mm^2})}{4} = 6.651\,\mathrm{mm^2}$. The diameter has three significant figures, so $\boxed{A = 6.65\,\mathrm{mm^2}}$.
EVALUATE: the diameter lies between 2.90 and $2.92\,\mathrm{mm}$. The 4 and the exponent 2 in the formula are exact and do not limit the figures.

A8. Accurate or precise?

Predict: X has readings that agree closely but sit away from $1.000\,\mathrm{kg}$; Y has scattered readings around $1.0\,\mathrm{kg}$.
IDENTIFY: accuracy is how close a value is likely to be to the true value; precision is how closely repeated measurements agree with one another.
SET UP: compare each student's mean with $1.000\,\mathrm{kg}$ and look at how closely each student's readings agree.
EXECUTE: X: mean $= \dfrac{1.0213 + 1.0215 + 1.0214}{3}\,\mathrm{kg} = 1.0214\,\mathrm{kg}$, which is $0.0214\,\mathrm{kg}$ (about 2%) too high. Y: mean $= \dfrac{0.9 + 1.1 + 1.0}{3}\,\mathrm{kg} = 1.0\,\mathrm{kg}$.
EVALUATE: X is more precise (the readings agree closely) but less accurate; Y is more accurate on average but much less precise. A good measurement is both.

A9. Order-of-magnitude estimate

Predict: a few billion.
IDENTIFY: target N, the total number of beats in 70 years at 70 beats per minute.
SET UP: $N =$ (beats per minute) $\times$ (minutes per hour) $\times$ (hours per day) $\times$ (days per year) $\times$ (years).
EXECUTE:
$N = 70 \times 60 \times 24 \times 365 \times 70$.
$70 \times 60 \times 24 \times 365 = 3.68 \times 10^7$ beats per year.
$N = 3.68 \times 10^7 \times 70 = 2.6 \times 10^9 \approx \boxed{3 \times 10^9}$.
EVALUATE: order of magnitude 10^9, in line with the prediction. Using 60 or 80 beats per minute would change the answer by about 15%, within the factor of a few that is the usual accuracy of such an estimate, and it does not change the power of ten.

A10. Density of a block

Predict: a dense metal, roughly $10\,\mathrm{g/cm^3}$.
IDENTIFY: density $\rho = m/V$.
SET UP: $V = lwh$; $1\,\mathrm{g} = 10^{-3}\,\mathrm{kg}$ and $1\,\mathrm{cm^3} = 10^{-6}\,\mathrm{m^3}$. Significant figures: 2.50 (3), 4.0 (2), 12.35 (4), m (3), so the result has 2.
EXECUTE:
$V = (2.50\,\mathrm{cm})(4.0\,\mathrm{cm})(12.35\,\mathrm{cm}) = 123.5\,\mathrm{cm^3}$ (keep the extra digits).
$\rho = \dfrac{1.20 \times 10^3\,\mathrm{g}}{123.5\,\mathrm{cm^3}} = 9.717\,\mathrm{g/cm^3} = \boxed{9.7\,\mathrm{g/cm^3}}$.
$\rho = 9.717\,\dfrac{\mathrm{g}}{\mathrm{cm^3}} \times \dfrac{10^{-3}\,\mathrm{kg}}{1\,\mathrm{g}} \times \dfrac{1\,\mathrm{cm^3}}{10^{-6}\,\mathrm{m^3}} = 9.717 \times 10^3\,\mathrm{kg/m^3} = \boxed{9.7 \times 10^3\,\mathrm{kg/m^3}}$.
EVALUATE: close to the prediction. $1\,\mathrm{g/cm^3} = 10^3\,\mathrm{kg/m^3}$, so the two answers agree. Rounding was done once, at the end.

C. Exam-style practice (multiple choice)

C1

A cyclist travels at $54\,\mathrm{km/h}$. Her speed in $\mathrm{m/s}$ is
(a) 0.015 (b) 15 (c) 54 (d) 194

C2

The number 0.04050 has how many significant figures?
(a) 2 (b) 3 (c) 4 (d) 5

C3

A rectangle is $3.2\,\mathrm{cm}$ by $1.25\,\mathrm{cm}$. Its area, with the correct number of significant figures, is
(a) $4\,\mathrm{cm}^2$ (b) $4.0\,\mathrm{cm}^2$ (c) $4.00\,\mathrm{cm}^2$ (d) $4.000\,\mathrm{cm}^2$

C4

One cubic metre equals
(a) $10^2\,\mathrm{cm}^3$ (b) $10^3\,\mathrm{cm}^3$ (c) $10^6\,\mathrm{cm}^3$ (d) $10^9\,\mathrm{cm}^3$

C5

A digital clock shows the time to the second, repeats its readings consistently, but runs 5 minutes slow. Its readings are
(a) precise and accurate (b) precise but not accurate (c) accurate but not precise (d) neither

C6

Which equation is dimensionally inconsistent?
(a) $F = ma$ (b) $E = mc^2$ (c) $v = at^2$ (d) $d = vt$

Answers to Part C

C1 (b): $54 \div 3.6 = 15\,\mathrm{m/s}$. C2 (c): leading zeros do not count, giving 4, 0, 5, 0. C3 (b): $3.2 \times 1.25 = 4.00$, and 3.2 has two figures, so $4.0\,\mathrm{cm}^2$. C4 (c): $1\,\mathrm{m} = 10^2\,\mathrm{cm}$, so $1\,\mathrm{m}^3 = (10^2)^3\,\mathrm{cm}^3 = 10^6\,\mathrm{cm}^3$. C5 (b): the readings are consistent and given to the second, so the clock is precise, but it is wrong by minutes, so it is not accurate. C6 (c): at^2 has units m, but v is in $\mathrm{m/s}$.